Kali Linux Setup

Objectives

This is part of lab setup, in this section Kali Linux will be installed. It will be mainly used to simulate attacker or offensive security. Kali Linux itself contains huge variety of tools that allows to perform penetration testing and ethical hacking. The following objectives are:

- Download Kali Linux
- Configure Virtual Machine before installing it
- Update the systems packages using update & upgrade
- Configure lab network

VM Overview

Kali Linux 2025.2

Purpose	Attacker box / Offensive Security
Base OS	Debian-based Linux
Installed On	VirtualBox 7.1.4
Resources Allocated	2 CPUs, 2 GB RAM, 30 GB Disk
Network Mode	Internal Network (LabNet)

Installation Process

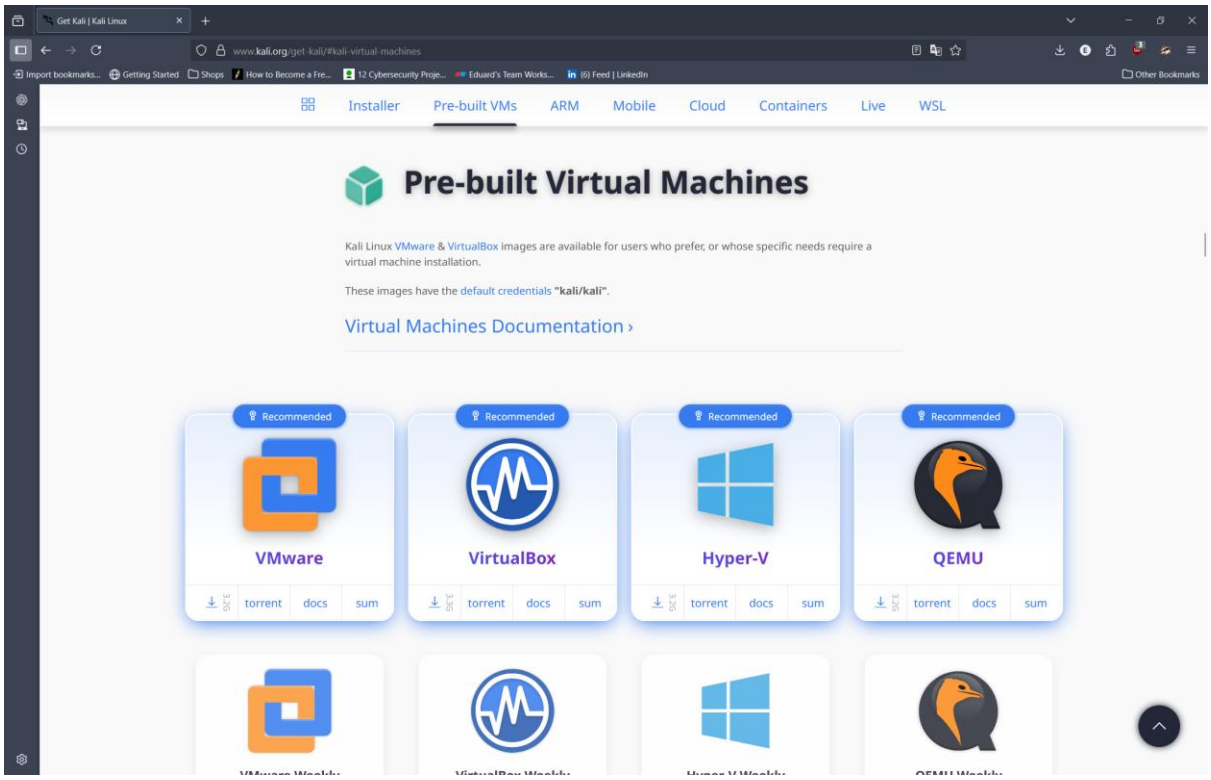
ISO Download

- Source: https://www.kali.org/get-kali/#kali-virtual-machines
- VIM File: kali-linux-2025.2-virtualbox-amd64.vim

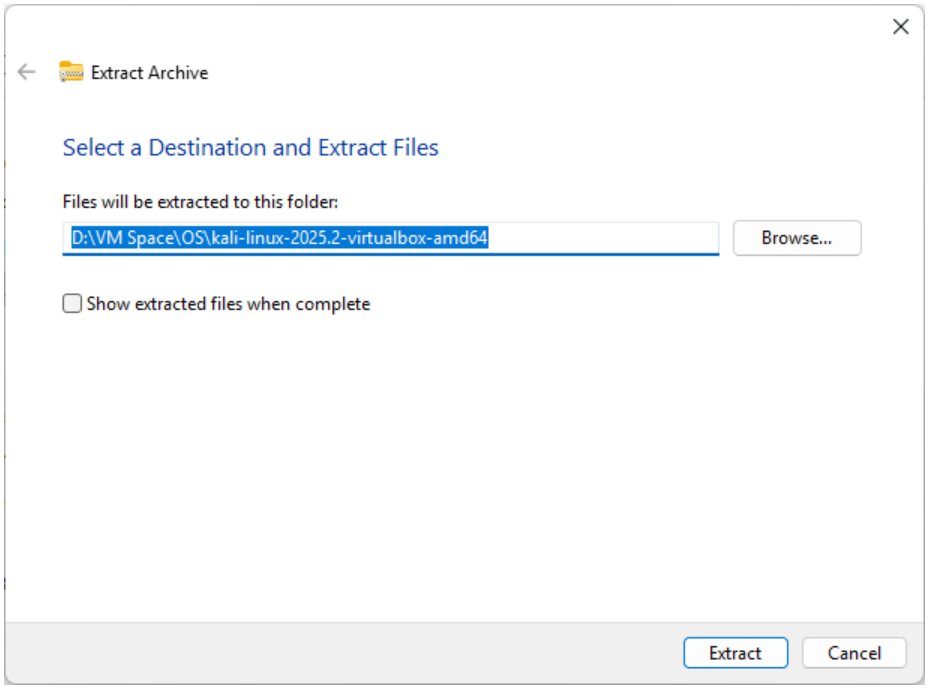
Virtual Machine Configuration

- Hypervisor: VirtualBox
- Name: Kali_Linux_2025.2
- Type: Linux → Debian (64-bit)
- Base Memory: 2048 MB
- CPUs: 2
- Network Adaptor: NAT (Later will be changed to Internal Adaptor)
- Hard Disk: 30 GB VDI (dynamically allocated)

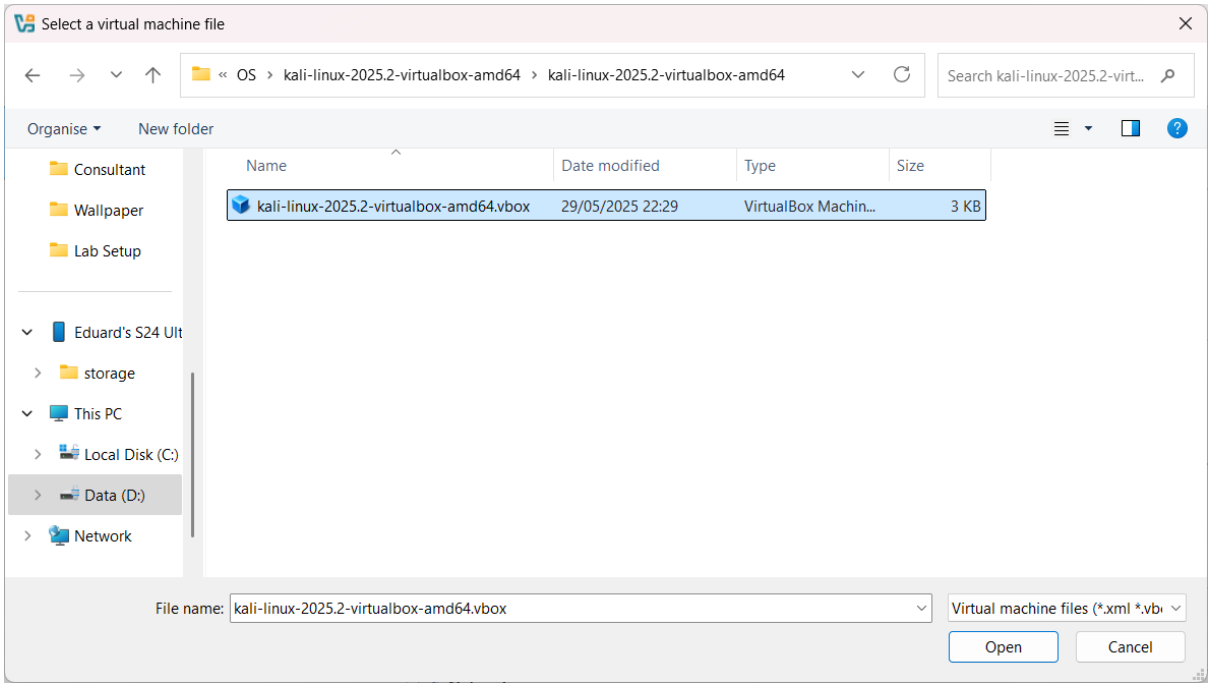
Screenshots



Kali Linux can be downloaded from the following link that was provided on previous page. Here it you can download the correct file for your system.

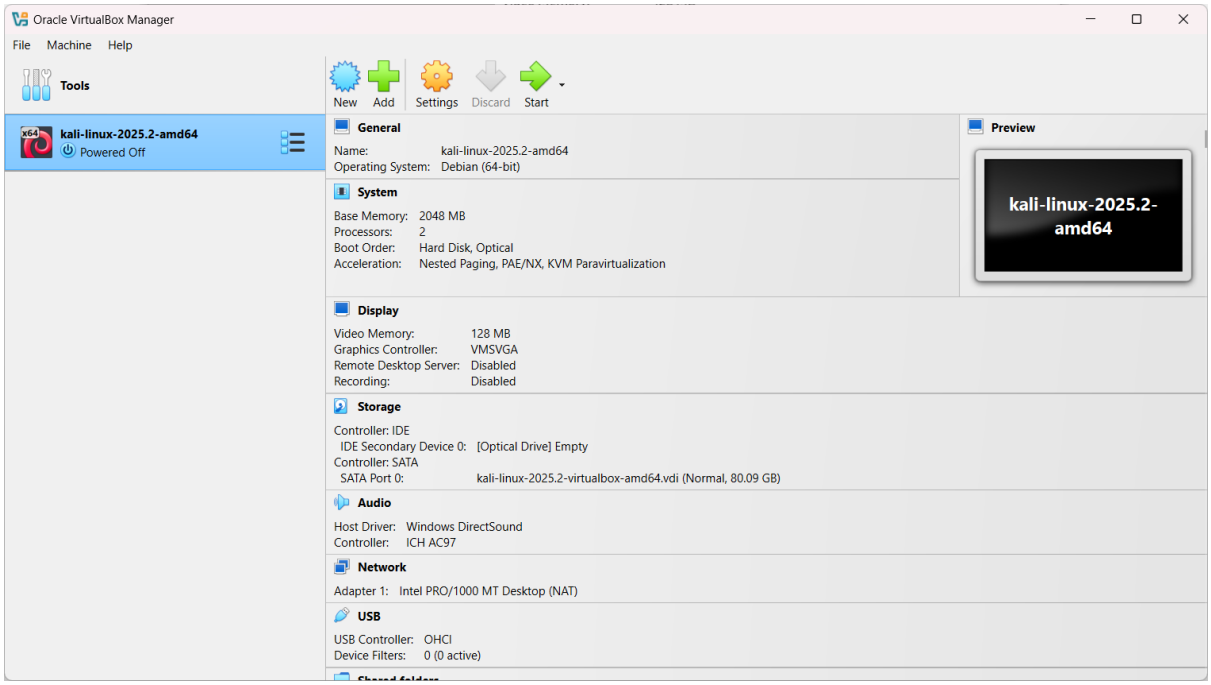


Once its downloaded, extract all the files into your chosen directory.

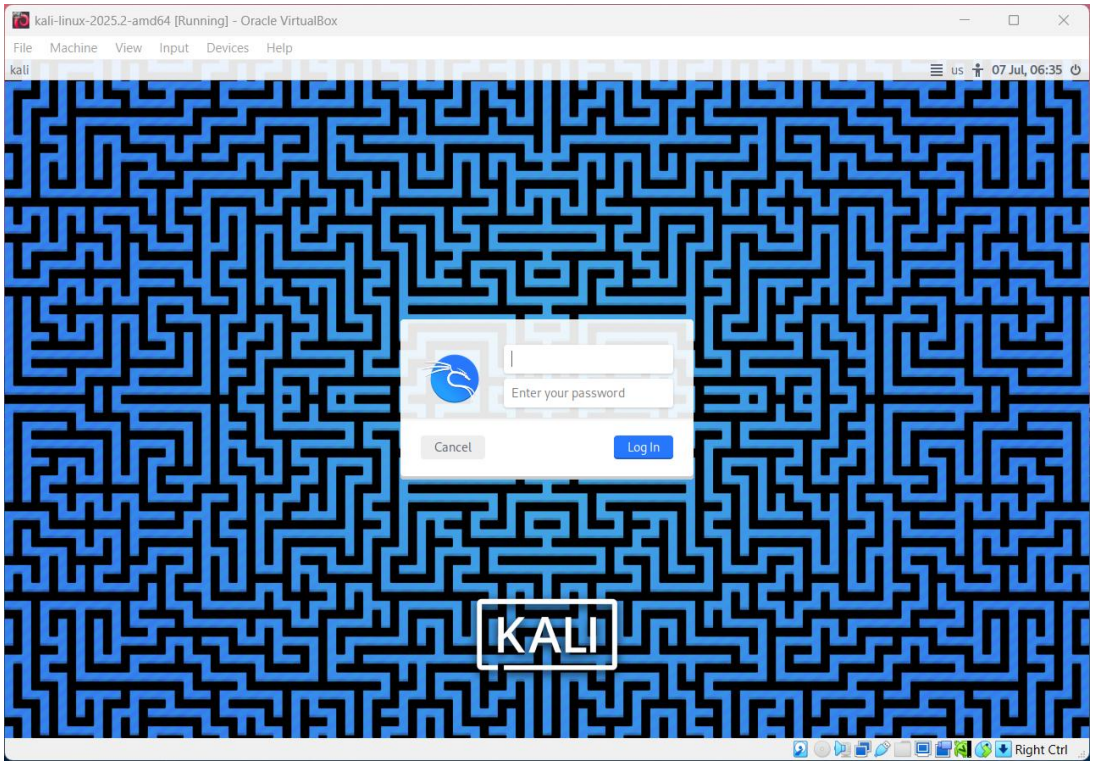


In your hypervisor choose add option and select the file that was extracted from the downloaded zip file (Only works if you downloaded a file for your specific hypervisor).

If you downloaded ISO image then use “New” option and select your VM file.

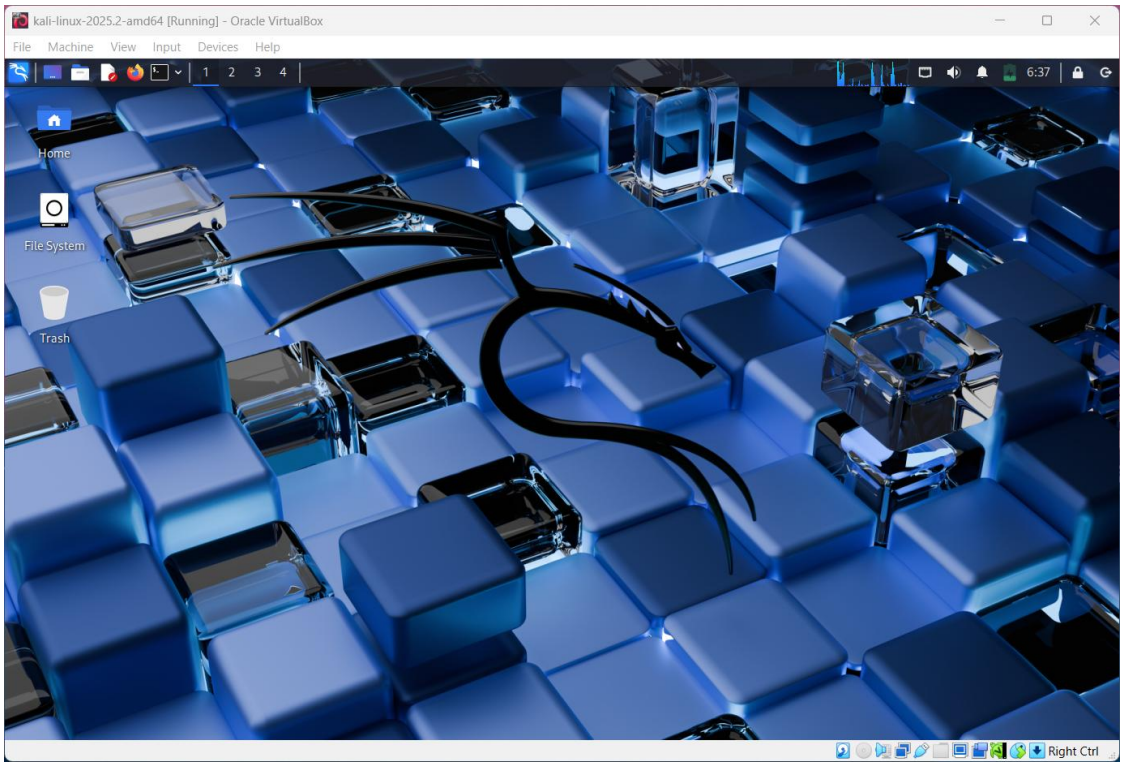


For this VM I will set base memory to 2GB and give 2 processors to it. This should be enough to run this machine smoothly.



The preinstalled machines for hypervisors come with a default username “Kali” and password “Kali”.

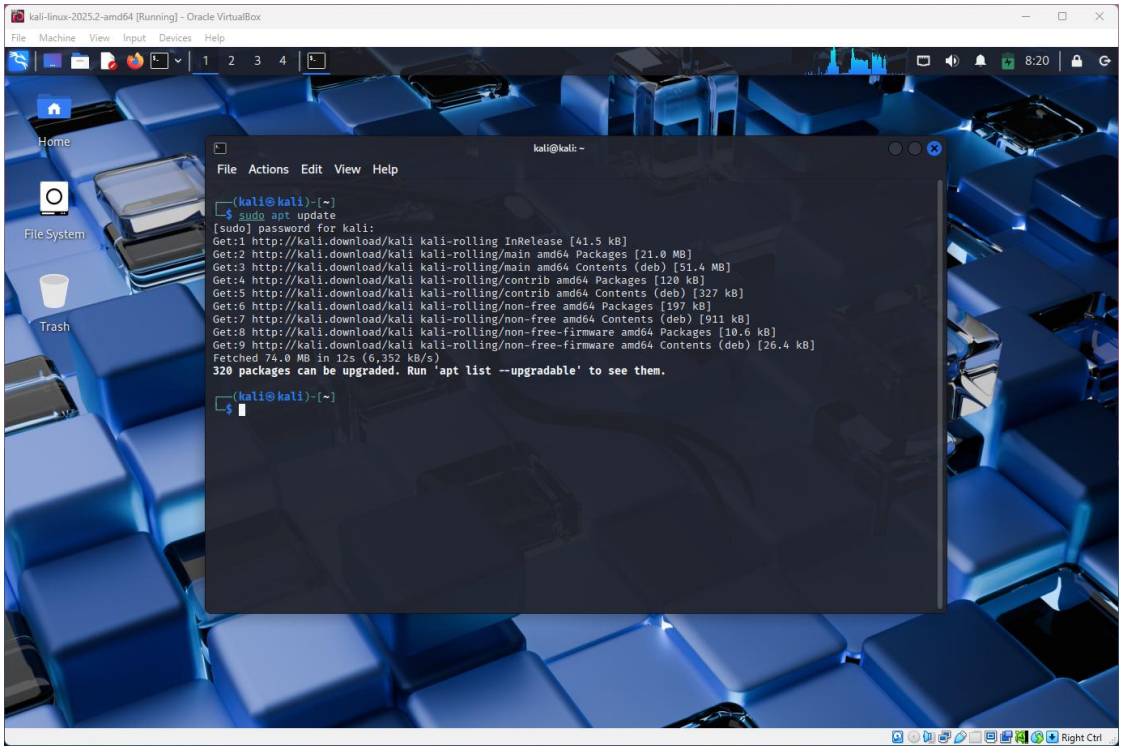
The machine that is installed through ISO file will require you to enter your chosen username and password during the installation process.



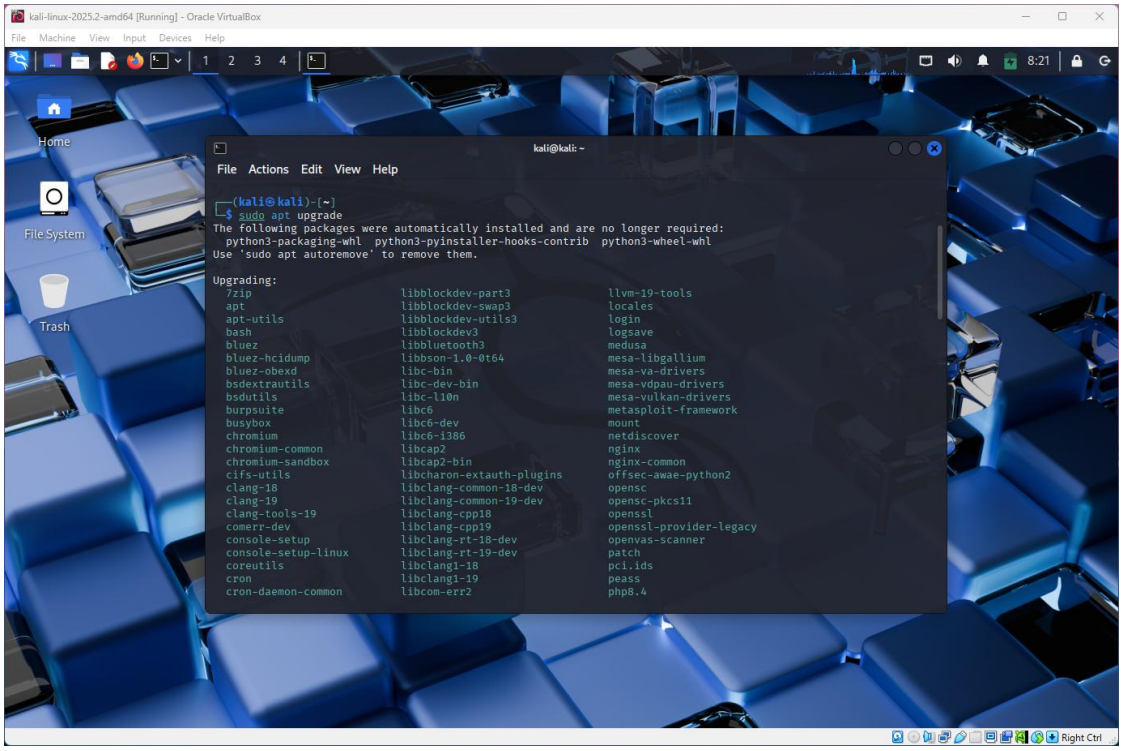
If everything done right, you should be able to log into the machine.

Post Installation Configuration

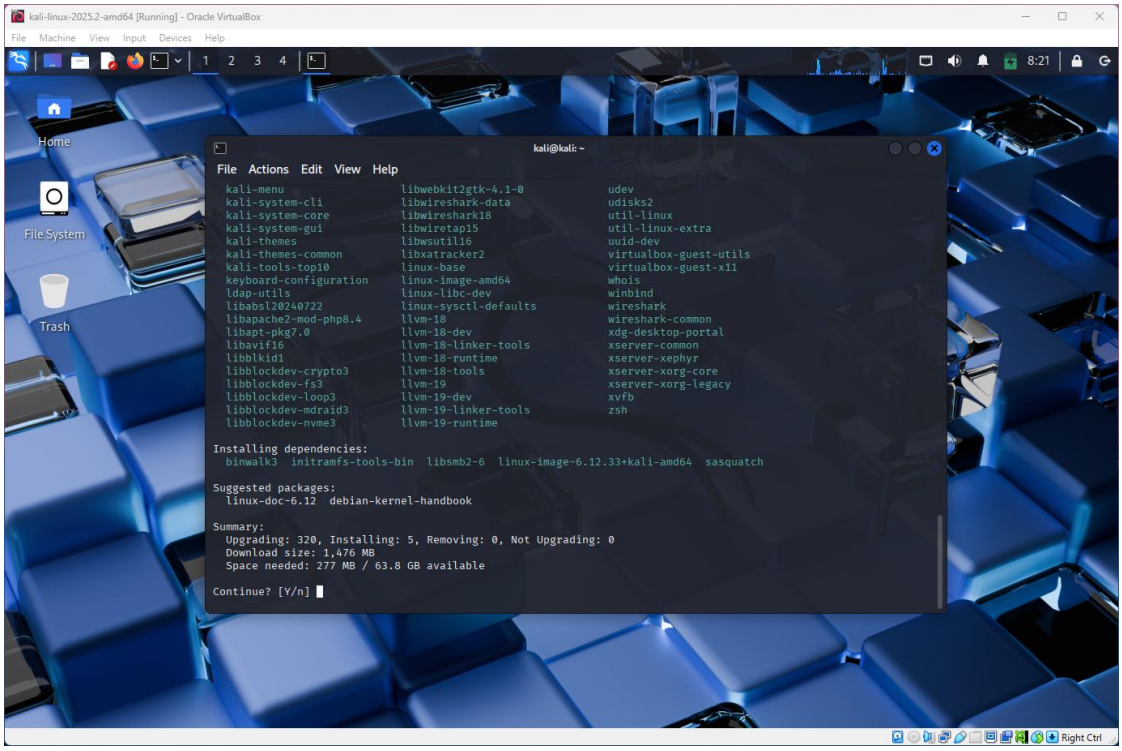
Update & Upgrade



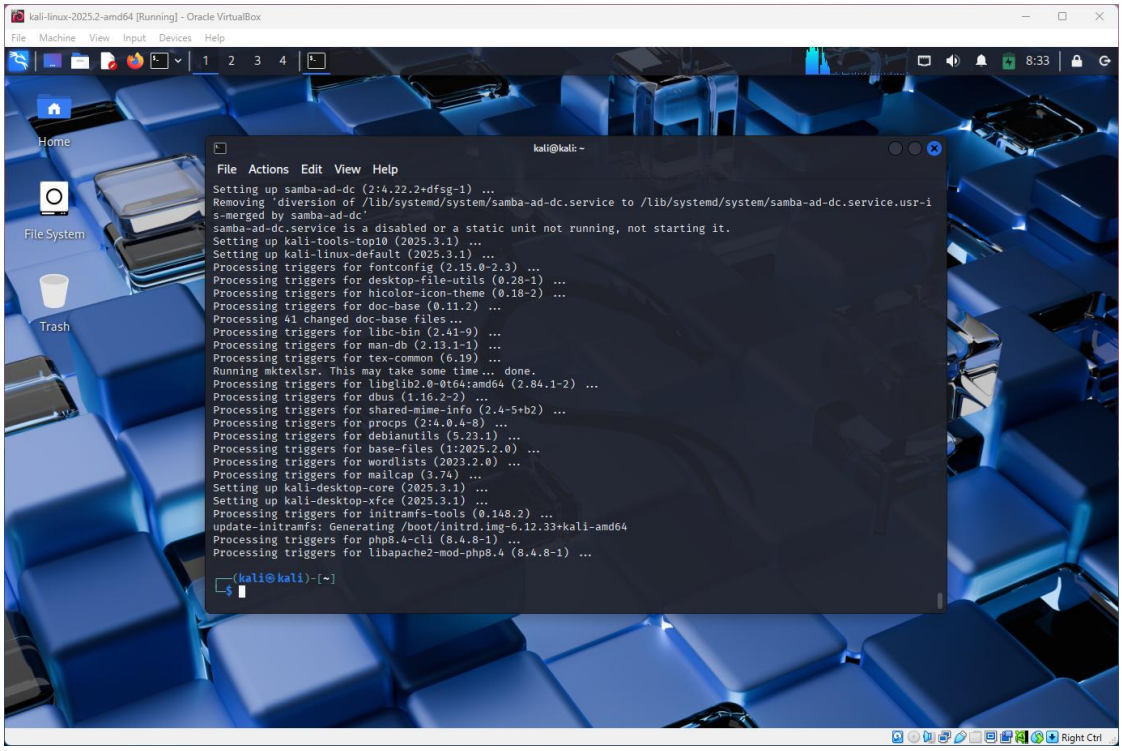
To get the latest packages for Kali Linux, or any Linux distributions, use “apt update”. However, it will not install them.



Once the packages have been fetch, they can be installed by using “apt upgrade”.

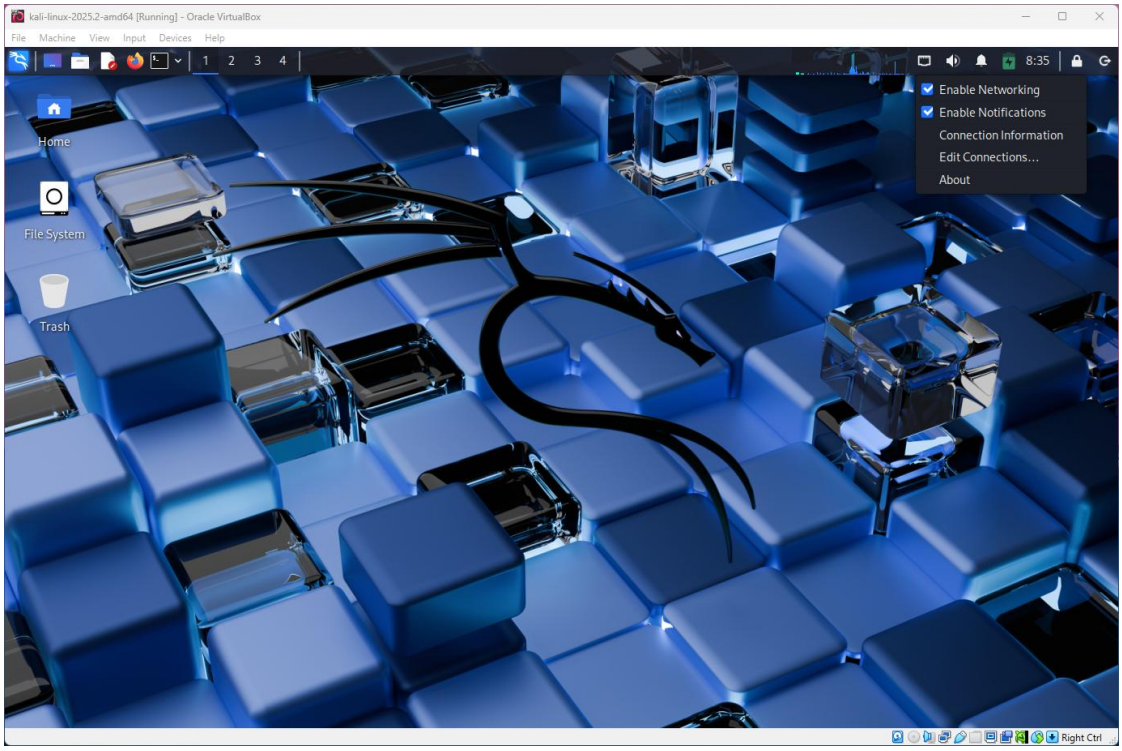


Once you run the command it will list all the packages that will be installed, and it will prompt you to continue or not. Enter “y” to continue.

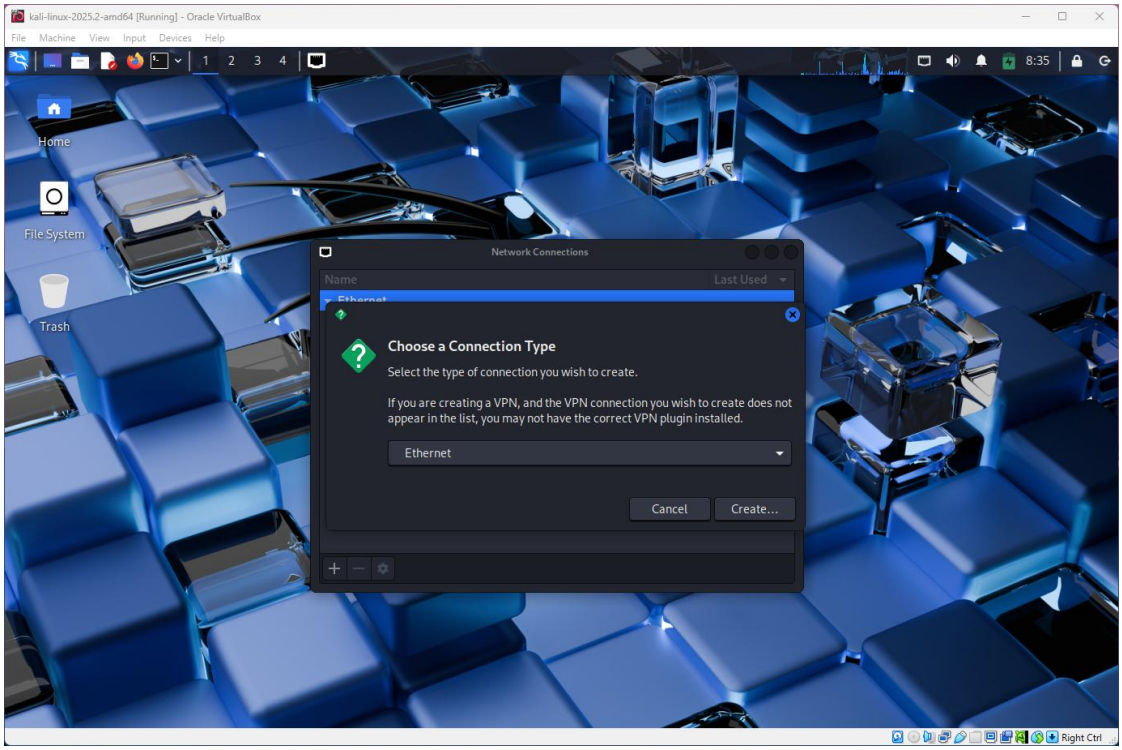


If the machine is not constantly updated, it may take a long time to install all the packages.

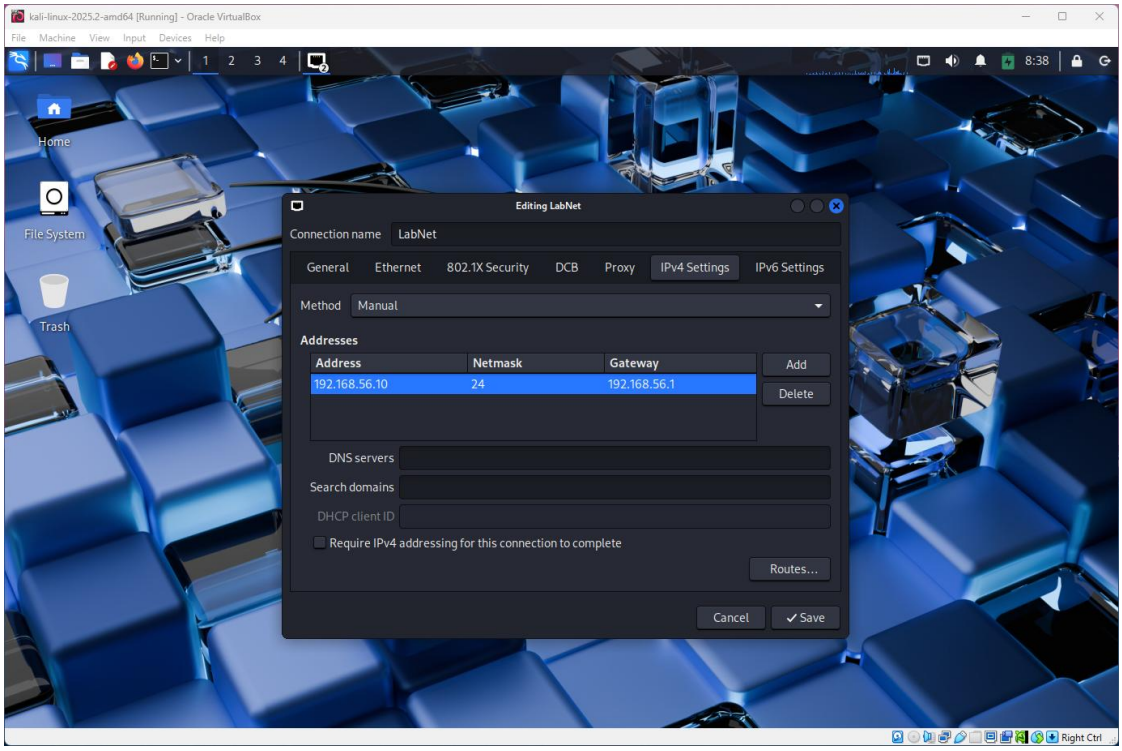
Static IP Configuration



In the task bar on the left side there is an icon that looks like ethernet port. Left click on it and press on edit connections.

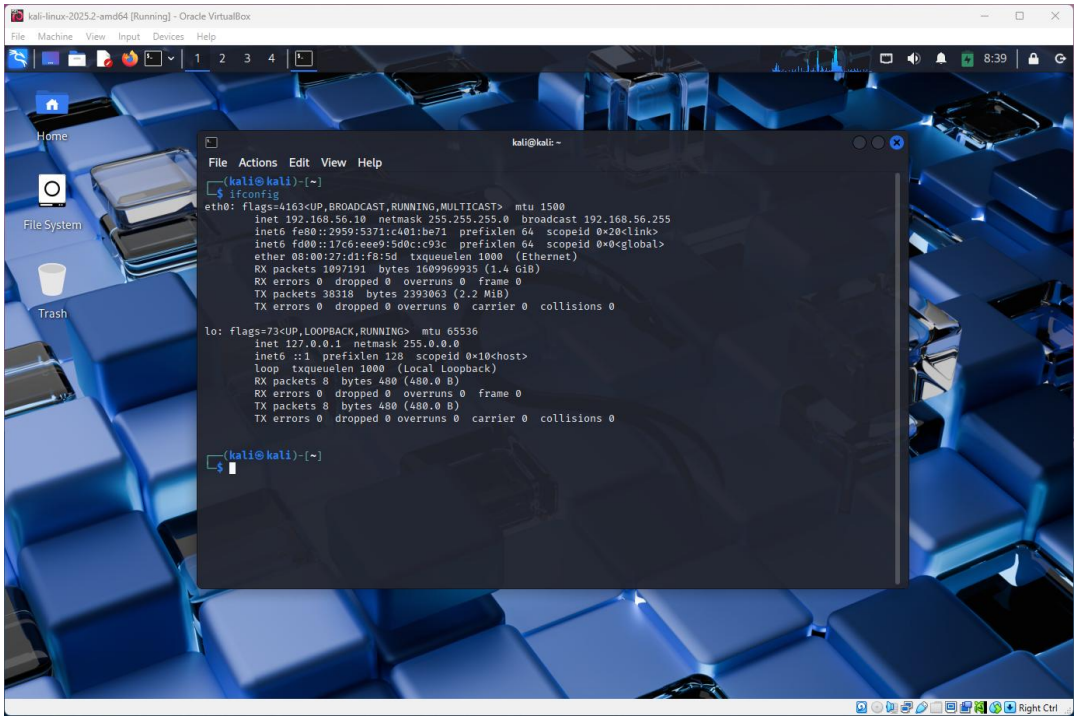


On the bottom press plus sign to create a new connection type. Select ethernet and press continue. Then select the newly created connection type and press on gear icon to edit the connection.

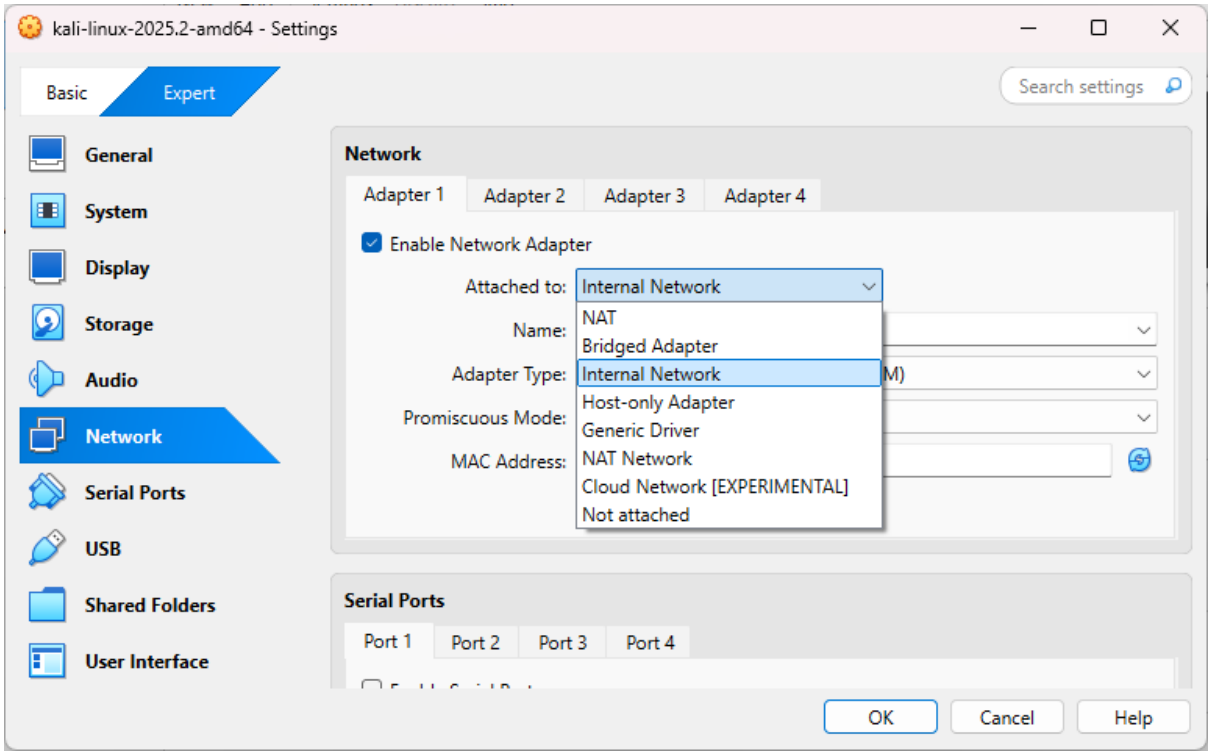


In here change the connection name to whatever suits you the best, in my case its “LabNet”. Then go to IPv4 Settings, before you do anything change method from DHCP to manual (for static IP address.)

Then add IP address, in my case its 197.168.56.10. The netmask is 24 and set gateway to the first usable IP address in that range (there is no rule for it, but it’s a default way).



Then save the configuration and select that connection type. To check if IP address is correct, we can use “ifconfig” command. In my case it was successfully changed.



To allow your machine to communicate with any other machine in your lab network you should select either Internal Network (this option will remove access to the internet) or Bridge Adaptor (allowing for communication in local network and outside one).

NAT option is a default one and allows you to access the internet and other services. However, it will not allow your machine to communicate with other machines in the lab network.

Future Plans / Next Steps

Since the Kali Linux is configured, it will be used for future labs to try to exploit any weaknesses to the systems I will use in my Lab environment.

Also, it contains a huge number of tools that I should learn about and practice.

References

Kali Linux Documentation: https://www.kali.org/docs/virtualization/import-premade-virtualbox/